



**Database System Final Project**  
**(Car Showroom Management System)**  
**(BS-SE-F23 Morning)**  
**INSTRUCTOR: SIR ASIF SOHAIL**

**MEMBERS:**

- **Abdul Mateen Butt (BSEF23M039)**
- **Haroon Ashraf Chodary (BSEF23M010)**
- **Muhammad Abdul Kareem (BSEF23M014)**
- **Muhammad Abdul Saboor (BSEF23M015)**

# *(Car Showroom Management System)*

## *(System Introduction)*

### *1. Working*

*The Car Showroom Management System automates and streamlines the operations of a car showroom. It manages customer information, car inventory, bookings, test drives, sales transactions, warranties, and insurance. By simplifying processes, the system improves business efficiency, reduces manual workload, and enhances customer satisfaction. It also supports decision-making through real-time insights and comprehensive reporting capabilities.*

### *2. Key Features of the System*

- Maintains detailed records of customers and their interactions with the showroom.*
- Manages car inventory with attributes like model, price, availability, and specifications.*
- Processes sales transactions, including customer details, payment methods, and delivery schedules.*
- Generates reports based on sales, inventory, and other key metrics to support strategic decision-making.*

### *3. Description*

*This system stores data in relational tables with keys and constraints, ensuring consistency. Each table represents a key business object such as a customer, a car, or a sale. Relationships are managed using foreign keys, ensuring accurate record linkage across the database.*

### *4. Purpose*

*The purpose of this system is to simplify operations, provide accurate record-keeping, and enhance customer satisfaction. It supports business growth by automating tasks and offering insights into performance, inventory, and customer engagement.*



# *(Entities and Their Relations)*

## 1. Entities and Attributes

### 1. Car

- *Primary key: CarID*
- *Attributes: Make, Model, Year, Colour, EngineType, Transmission, Mileage, Price, VIN, Status*

### 2. Customer

- *Primary key: CustomerID*
- *Attributes: Name, Phone, Email, Address*

### 3. Salesperson

- *Primary key: SalespersonID*
- *Attributes: Name, Phone, Email, HireDate, Position, Salary*

### 4. Sale

- *Primary key: SaleID*
- *Attributes: DateOfSale, FinalPrice, PaymentMethod, CarID (Foreign Key), CustomerID (Foreign Key), SalespersonID (Foreign Key)*

### 5. TestDrive

- *Primary key: TestDriveID*
- *Attributes: ScheduledDate, Duration, Feedback, CarID (Foreign Key), CustomerID (Foreign Key), SalespersonID (Foreign Key)*

### 6. ShowRoom

- *Primary key: ShowroomID*

- *Attributes: Name, Location, Phone, AdminID*

## **7. Inventory**

- *Primary key: InventoryID*
- *Attributes: ItemName, Quantity, Category, ShowroomID (Foreign Key)*

## **8. Staff**

- *Primary key: StaffID*
- *Attributes: Name, Role, ShowroomID (Foreign Key), Contact*

## **9. Payroll**

- *Primary key: PayID*
- *Attributes: Salary, PayDate, StaffID (Foreign Key)*

## **10. Administration**

- *Primary key: AdminID*
- *Attributes: Name, Contact, Role, ShowroomID (Foreign Key)*

## **11. Insurance**

- *Primary key: InsuranceID*
- *Attributes: CarID (Foreign Key), CustomerID (Foreign Key), InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount*

## **12. Booking**

- *Primary key: BookingID*
- *Attributes: CustomerID (Foreign Key), CarID (Foreign Key), BookingDate, BookingStatus, AdvanceAmount*

## **13. Warranty**

- *Primary key: WarrantyID*

- *Attributes: CarID (Foreign Key), WarrantyStartDate, WarrantyEndDate, CoverageDetails*

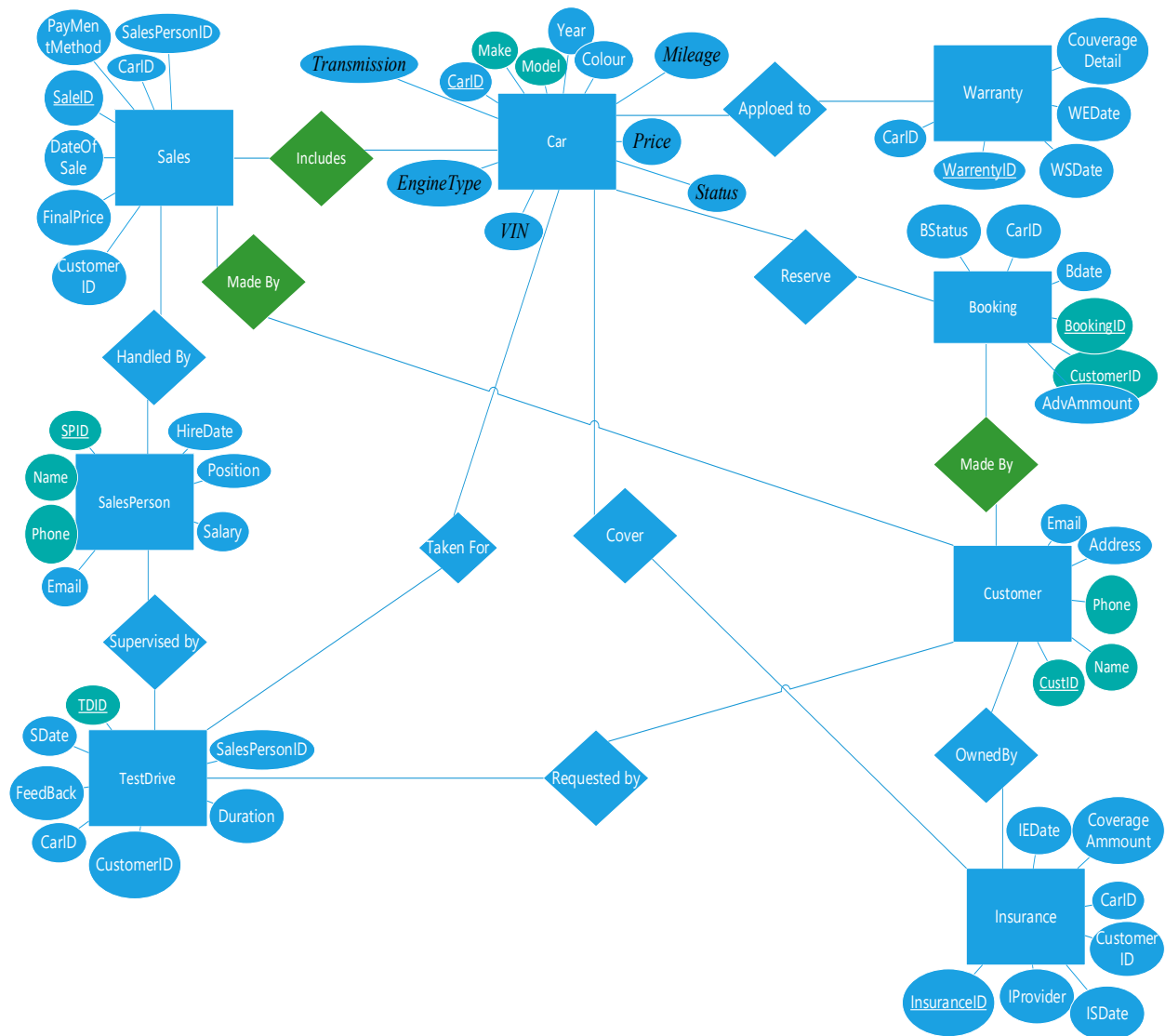
## **2. Relationships**

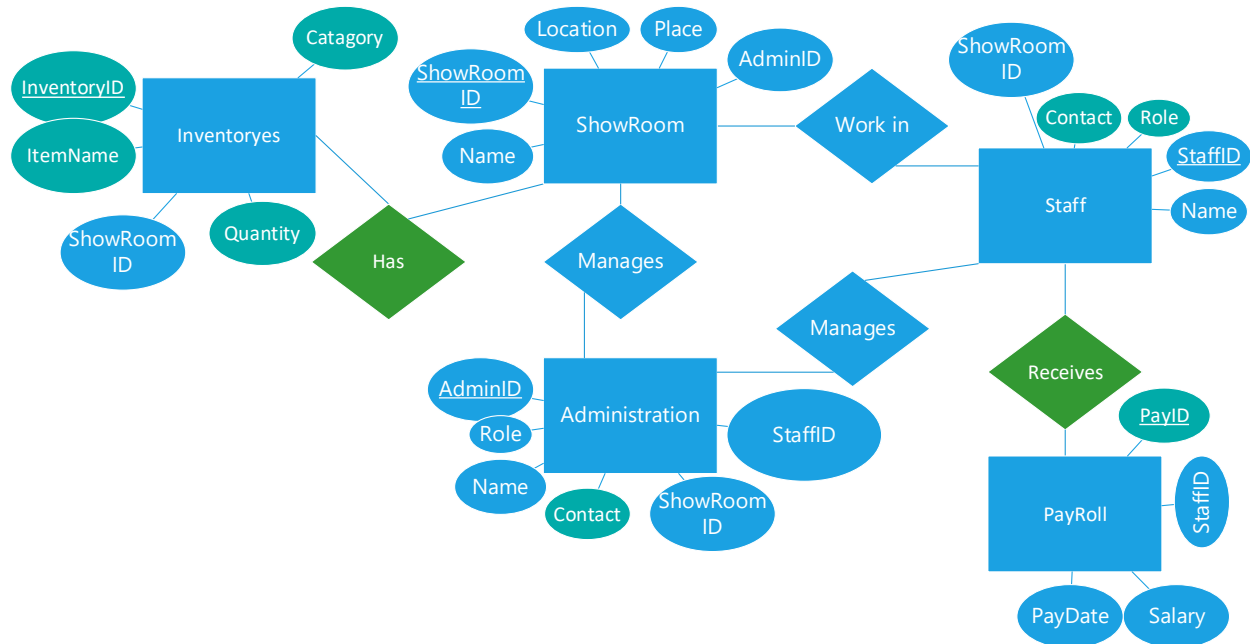
- 1. Sale and Car**  
*Relationship: Includes*
- 2. Sales and Customer**  
*Relationship: Made by*
- 3. Sales and Salesperson**  
*Relationship: Handled by*
- 4. Booking and Customer**  
*Relationship: Made by*
- 5. Booking and Car**  
*Relationship: Reserves*
- 6. TestDrive and Customer**  
*Relationship: Requested by*
- 7. TestDrive and Car**  
*Relationship: Taken for*
- 8. TestDrive and Salesperson**  
*Relationship: Supervised by*
- 9. Warranty and Car**  
*Relationship: Applies to*
- 10. Insurance and Car**  
*Relationship: Covers*
- 11. Insurance and Customer**  
*Relationship: Owned by*
- 12. Payroll and Staff**  
*Relationship: Receives*
- 13. Staff and Showroom**  
*Relationship: Works in*
- 14. Administration and Staff**  
*Relationship: Manages*

15. *Administration and Showroom*  
Relationship: *Manages*

16. *Inventory and Showroom*  
Relationship: *has*

## (Entity Relationship Diagram)





## *(System Design and Functionalities)*

### *1. Database Design*

- *Developed using both top-down and bottom-up design approaches.*
- *Includes an Entity Relationship Diagram (ERD) to visually represent relationships between entities such as Customers, Cars, Sales, and Staff.*
- *Tables are defined with attributes, constraints, and relational keys to ensure consistency and data integrity*

### *2. Advanced Functionalities*

- *Custom Views: Enable simplified access to frequently needed information, such as available cars or customer bookings.*
- *Query-Based Reports: Generate real-time insights including total sales per month, top-performing salespersons, and car availability by showroom.*

# *(Relational Schema)*

## 1. Top-Down-Approch

### 1. Car

- *(CarID (PK), Make, Model, Year, Color, EngineType, Transmission, Mileage, Price, VIN, Status)*

### 2. Customer

- *(CustomerID (PK), Name, Phone, Email, Address)*

### 3. Salesperson

- *(SalespersonID (PK), Name, Phone, Email, HireDate, Position, Salary)*

### 4. Sale

- *(SaleID (PK), DateOfSale, FinalPrice, PaymentMethod, CustomerID (FK referencing Customer), CarID (FK referencing Car), SalespersonID (FK referencing Salesperson))*

### 5. TestDrive

- *(TestDriveID (PK), ScheduledDate, Duration, Feedback, CarID (FK referencing Car), CustomerID (FK referencing Customer), SalespersonID (FK referencing Salesperson))*

### 6. ShowRoom

- *(ShowroomID (PK), Name, Location, Phone, AdminID)*

### 7. Inventory

- *(InventoryID (PK), ItemName, Quantity, Category, ShowroomID (FK referencing Showroom))*



## 8. Staff

- (*StaffID (PK), Name, Role, Contact, ShowroomID (FK referencing Showroom)*)

## 9. Payroll

- (*PayID (PK), Salary, PayDate, StaffID (FK referencing Staff)*)

## 10. Administration

- (*AdminID (PK), Name, Contact, Role, ShowroomID (FK referencing Showroom)*)

## 11. Insurance

- (*InsuranceID (PK), InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount, CarID (FK referencing Car), CustomerID (FK referencing Customer)*)

## 12. Booking

- (*BookingID (PK), BookingDate, BookingStatus, AdvanceAmount, CustomerID (FK referencing Customer), CarID (FK referencing Car)*)

## 13. Warranty

- (*WarrantyID (PK), WarrantyStartDate, WarrantyEndDate, CoverageDetails, CarID (FK referencing Car)*)

## 2. Bottom-Up-Approach

Based on the following criteria, we will do our Bottom Up Approach:

1. A customer may purchase multiple cars (CarID group).
2. A customer may take multiple test drives (TestDriveID group).
3. A customer might have multiple bookings (BookingID group).
4. Showrooms have multiple staff members.
5. Admins could also have multiple contacts/roles? (though that one's fuzzy).
6. Salespersons might be assigned to multiple sales.

## 1. 0NF

*R ( CustomerID, Name, Phone, Email, Address, CarID, Make, Model, Year, Color, EngineType, Transmission, Mileage, Price, VIN, Status, SaleID, DateOfSale, FinalPrice, PaymentMethod, SalespersonID, SalespersonName, SalespersonPhone, SalespersonEmail, HireDate, Position, Salary, TestDriveID, ScheduledDate, Duration, Feedback, ShowroomID, ShowroomName, Location, ShowroomPhone, ManagerName, InventoryID, ItemName, Quantity, Category, StaffID, StaffName, StaffRole, StaffContact, PayID, SalaryAmount, PayDate, AdminID, AdminName, AdminContact, AdminRole, InsuranceID, InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount, BookingID, BookingDate, BookingStatus, AdvanceAmount, WarrantyID, WarrantyStartDate, WarrantyEndDate, CoverageDetails )*

## 2. 1NF

### 1. **Customer**

**Customer** (CustomerID, Name, Phone, Email, Address)

### 2. **Car**

**Car** (CarID, Make, Model, Year, Colour, EngineType, Transmission, Mileage, Price, VIN, Status, WarrantyID, WarrantyStartDate, WarrantyEndDate, CoverageDetails)

### 3. **Salesperson**

**Salesperson** (SalespersonID, SalespersonName, SalespersonPhone, SalespersonEmail, HireDate, Position, Salary)

### 4. **Sale**

**Sale** (SaleID, CarID, CustomerID, DateOfSale, FinalPrice, PaymentMethod, SalespersonID)

### 5. **TestDrive**

**TestDrive** (TestDriveID, CustomerID, CarID, ScheduledDate, Duration, Feedback)

### 6. **Showroom**

**Showroom** (ShowroomID, ShowroomName, Location, ShowroomPhone, AdminID)

### 7. **Inventory**

**Inventory** (InventoryID, ShowroomID, ItemName, Quantity, Category)

8. **Staff**  
**Staff** (*StaffID, ShowroomID, StaffName, StaffRole, StaffContact, PayID, SalaryAmount, PayDate*)
9. **Administration**  
**Admin** (*AdminID, AdminName, AdminContact, AdminRole*)
10. **Insurance**  
**Insurance** (*InsuranceID, CarID, InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount*)
11. **Booking**  
**Booking** (*BookingID, CustomerID, CarID, BookingDate, BookingStatus, AdvanceAmount*)

### 3. 2NF

Already in 2NF — there are no partial dependencies in the above 1NF relations.

### 4. 3NF

- 1) **Customer**  
(*CustomerID, Name, Phone, Email, Address*)
- 2) **Car**  
(*CarID, Make, Model, Year, Colour, EngineType, Transmission, Mileage, Price, VIN, Status*)
- 3) **Salesperson**  
(*SalespersonID, Name, Phone, Email, HireDate, Position, Salary*)
- 4) **Sale**  
(*SaleID, DateOfSale, FinalPrice, PaymentMethod, CustomerID, CarID, SalespersonID*)
- 5) **TestDrive**  
(*TestDriveID, ScheduledDate, Duration, Feedback, CarID, CustomerID, SalespersonID*)
- 6) **Showroom**  
(*ShowroomID, Name, Location, Phone, AdminID*)
- 7) **Inventory**  
(*InventoryID, ItemName, Quantity, Category, ShowroomID*)

### 8) **Staff**

(StaffID, Name, Role, Contact, ShowroomID, AdminID)

### 9) **Payroll**

(PayID, SalaryAmount, PayDate, StaffID)

### 10) **Administration**

(AdminID, Name, Contact, Role, ShowroomID)

### 11) **Insurance**

(InsuranceID, InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount, CarID, CustomerID)

### 12) **Booking**

(BookingID, BookingDate, BookingStatus, AdvanceAmount, CustomerID, CarID)

### 13) **Warranty**

(WarrantyID, WarrantyStartDate, WarrantyEndDate, CoverageDetails, CarID)

## 5. BCNF

Already in BCNF — in all relations, every determinant is a candidate key, so no violations exist.

## 6. 4NF

Already in 4NF — no relation contains attributes with multivalued dependencies not already tied to a candidate key.

## 7. 5NF

Already in 5NF — no relation needs to be decomposed into smaller projections to eliminate anomalies or preserve information.

# (Table Description)

## 1. Customer

| Attribute  | Data Type | Size | Constraints |
|------------|-----------|------|-------------|
| CustomerID | Number    | -    | Primary Key |
| Name       | Varchar2  | 30   | Not Null    |
| Phone      | Varchar2  | 15   | Not Null    |
| Email      | Varchar2  | 30   | -           |
| Address    | Varchar2  | 50   | Not Null    |

## 2. Car

| Attribute           | Data Type | Size | Constraints                            |
|---------------------|-----------|------|----------------------------------------|
| <b>CarID</b>        | Number    | -    | Primary Key                            |
| <b>Make</b>         | Varchar2  | 20   | Not Null                               |
| <b>Model</b>        | Varchar2  | 20   | Not Null                               |
| <b>Year</b>         | Number    | -    | -                                      |
| <b>Color</b>        | Varchar2  | 10   | -                                      |
| <b>EngineType</b>   | Varchar2  | 15   | -                                      |
| <b>Transmission</b> | Varchar2  | 10   | -                                      |
| <b>Mileage</b>      | Number    | -    | -                                      |
| <b>Price</b>        | Number    | -    | -                                      |
| <b>VIN</b>          | Varchar2  | 17   | Unique                                 |
| <b>Status</b>       | Varchar2  | 10   | Check(Status IN ('Available', 'Sold')) |

## 3. SalesPerson

| Attribute            | Data Type | Size | Constraints     |
|----------------------|-----------|------|-----------------|
| <b>SalespersonID</b> | Number    | -    | Primary Key     |
| <b>Name</b>          | Varchar2  | 30   | Not Null        |
| <b>Phone</b>         | Varchar2  | 15   | Not Null        |
| <b>Email</b>         | Varchar2  | 30   | -               |
| <b>HireDate</b>      | Date      | -    | Default SYSDATE |
| <b>Position</b>      | Varchar2  | 20   | -               |
| <b>Salary</b>        | Number    | -    | -               |

## 4. Sale

| Attribute            | Data Type | Size | Constraints                 |
|----------------------|-----------|------|-----------------------------|
| <b>SaleID</b>        | Number    | -    | Primary Key                 |
| <b>DateOfSale</b>    | Date      | -    | Default SYSDATE             |
| <b>FinalPrice</b>    | Number    | -    | -                           |
| <b>PaymentMethod</b> | Varchar2  | 15   | -                           |
| <b>CustomerID</b>    | Number    | -    | Foreign Key (CustomerID)    |
| <b>CarID</b>         | Number    | -    | Foreign Key (CarID)         |
| <b>SalespersonID</b> | Number    | -    | Foreign Key (SalespersonID) |

## 5. TestDrive

| Attribute            | Data Type | Size | Constraints |
|----------------------|-----------|------|-------------|
| <b>TestDriveID</b>   | Number    | -    | Primary Key |
| <b>ScheduledDate</b> | Date      | -    | -           |

|                      |          |     |                             |
|----------------------|----------|-----|-----------------------------|
| <b>Duration</b>      | Varchar2 | 5   | -                           |
| <b>Feedback</b>      | Varchar2 | 100 | -                           |
| <b>CarID</b>         | Number   | -   | Foreign Key (CarID)         |
| <b>CustomerID</b>    | Number   | -   | Foreign Key (CustomerID)    |
| <b>SalespersonID</b> | Number   | -   | Foreign Key (SalespersonID) |

## 6. ShowRoom

| Attribute         | Data Type | Size | Constraints                    |
|-------------------|-----------|------|--------------------------------|
| <b>ShowroomID</b> | Number    | -    | Primary Key                    |
| <b>Name</b>       | Varchar2  | 30   | Not Null                       |
| <b>Location</b>   | Varchar2  | 50   | Not Null                       |
| <b>Phone</b>      | Varchar2  | 15   | Not Null                       |
| <b>AdminID</b>    | Number    | -    | Foreign Key (AdministrationID) |

## 7. Inventory

| Attribute          | Data Type | Size | Constraints              |
|--------------------|-----------|------|--------------------------|
| <b>InventoryID</b> | Number    | -    | Primary Key              |
| <b>ItemName</b>    | Varchar2  | 25   | Not Null                 |
| <b>Quantity</b>    | Number    | -    | Check(Quantity >= 0)     |
| <b>Category</b>    | Varchar2  | 20   | Not Null                 |
| <b>ShowroomID</b>  | Number    | -    | Foreign Key (ShowroomID) |

## 8. Staff

| Attribute         | Data Type | Size | Constraints              |
|-------------------|-----------|------|--------------------------|
| <b>StaffID</b>    | Number    | -    | Primary Key              |
| <b>Name</b>       | Varchar2  | 30   | Not Null                 |
| <b>Role</b>       | Varchar2  | 20   | Not Null                 |
| <b>Contact</b>    | Varchar2  | 15   | Not Null                 |
| <b>ShowroomID</b> | Number    | -    | Foreign Key (ShowroomID) |

## 9. Payroll

| Attribute           | Data Type | Size | Constraints |
|---------------------|-----------|------|-------------|
| <b>PayID</b>        | Number    | -    | Primary Key |
| <b>SalaryAmount</b> | Number    | -    | -           |

|                |        |   |                       |
|----------------|--------|---|-----------------------|
| <b>PayDate</b> | Date   | - | Default SYSDATE       |
| <b>StaffID</b> | Number | - | Foreign Key (StaffID) |

### 10. Administration

| Attribute         | Data Type | Size | Constraints              |
|-------------------|-----------|------|--------------------------|
| <b>AdminID</b>    | Number    | -    | Primary Key              |
| <b>Name</b>       | Varchar2  | 30   | Not Null                 |
| <b>Contact</b>    | Varchar2  | 15   | Not Null                 |
| <b>Role</b>       | Varchar2  | 20   | Not Null                 |
| <b>ShowroomID</b> | Number    | -    | Foreign Key (ShowroomID) |

### 11. Insurance

| Attribute                 | Data Type | Size | Constraints              |
|---------------------------|-----------|------|--------------------------|
| <b>InsuranceID</b>        | Number    | -    | Primary Key              |
| <b>InsuranceProvider</b>  | Varchar2  | 30   | Not Null                 |
| <b>InsuranceStartDate</b> | Date      | -    | -                        |
| <b>InsuranceEndDate</b>   | Date      | -    | -                        |
| <b>CoverageAmount</b>     | Number    | -    | -                        |
| <b>CarID</b>              | Number    | -    | Foreign Key (CarID)      |
| <b>CustomerID</b>         | Number    | -    | Foreign Key (CustomerID) |

### 12. Booking

| Attribute            | Data Type | Size | Constraints                                                   |
|----------------------|-----------|------|---------------------------------------------------------------|
| <b>BookingID</b>     | Number    | -    | Primary Key                                                   |
| <b>BookingDate</b>   | Date      | -    | Default SYSDATE                                               |
| <b>BookingStatus</b> | Varchar2  | 20   | Check(BookingStatus IN ('Pending', 'Confirmed', 'Cancelled')) |
| <b>AdvanceAmount</b> | Number    | -    | -                                                             |
| <b>CustomerID</b>    | Number    | -    | Foreign Key (CustomerID)                                      |
| <b>CarID</b>         | Number    | -    | Foreign Key (CarID)                                           |

### 13. Warranty

| Attribute         | Data Type | Size | Constraints |
|-------------------|-----------|------|-------------|
| <b>WarrantyID</b> | Number    | -    | Primary Key |

|                   |          |     |                     |
|-------------------|----------|-----|---------------------|
| WarrantyStartDate | Date     | -   | -                   |
| WarrantyEndDate   | Date     | -   | -                   |
| CoverageDetails   | Varchar2 | 100 | -                   |
| CarID             | Number   | -   | Foreign Key (CarID) |

## (SQL Statements)

### 1. Create Statement

#### 1. Customer

```
CREATE TABLE Customer
(
    CustomerID NUMBER CONSTRAINT pk_customer PRIMARY KEY,
    Name VARCHAR2(30) CONSTRAINT nn_customer_name NOTNULL,
    Phone VARCHAR2(15) CONSTRAINT nn_customer_phone NOTNULL,
    Email VARCHAR2(30),
    Address VARCHAR2(50) CONSTRAINT nn_customer_address NOTNULL
);
```

#### 2. Car

```
CREATE TABLE Car
(
    CarID NUMBER CONSTRAINT pk_car PRIMARY KEY,
    Make VARCHAR2(20) CONSTRAINT nn_car_make NOT NULL,
    Model VARCHAR2(20) CONSTRAINT nn_car_model NOT NULL,
    Year NUMBER,
    Color VARCHAR2(10),
    EngineType VARCHAR2(15),
    Transmission VARCHAR2(10),
    Mileage NUMBER,
    Price NUMBER,
    VIN VARCHAR2(17) CONSTRAINT uq_car_vin UNIQUE,
    Status VARCHAR2(10) CONSTRAINT chk_car_status CHECK (Status IN
('Available', 'Sold'))
);
```

#### 3. Salesperson

```
CREATE TABLE Salesperson
(
    SalespersonID NUMBER CONSTRAINT pk_salesperson PRIMARY KEY,
```



```

    Name VARCHAR2(30) CONSTRAINT nn_sp_name NOT NULL,
    Phone VARCHAR2(15) CONSTRAINT nn_sp_phone NOT NULL,
    Email VARCHAR2(30),
    HireDate DATE DEFAULT SYSDATE,
    Position VARCHAR2(20),
    Salary NUMBER
);

```

#### 4. Sale

```

CREATE TABLE Sale
(
    SaleID NUMBER CONSTRAINT pk_sale PRIMARY KEY,
    DateOfSale DATE DEFAULT SYSDATE,
    FinalPrice NUMBER,
    PaymentMethod VARCHAR2(15),
    CustomerID NUMBER,
    CarID NUMBER,
    SalespersonID NUMBER,
    CONSTRAINT fk_sale_customer FOREIGN KEY (CustomerID) REFERENCES
Customer(CustomerID),
    CONSTRAINT fk_sale_car FOREIGN KEY (CarID) REFERENCES Car(CarID),
    CONSTRAINT fk_sale_salesperson FOREIGN KEY (SalespersonID)
REFERENCES Salesperson(SalespersonID)
);

```

#### 5. TestDrive

```

CREATE TABLE TestDrive
(
    TestDriveID NUMBER CONSTRAINT pk_testdrive PRIMARY KEY,
    ScheduledDate DATE,
    Duration VARCHAR2(5),
    Feedback VARCHAR2(100),
    CarID NUMBER,
    CustomerID NUMBER,
    SalespersonID NUMBER,
    CONSTRAINT fk_td_car FOREIGN KEY (CarID) REFERENCES
Car(CarID),
    CONSTRAINT fk_td_customer FOREIGN KEY (CustomerID)
REFERENCES Customer(CustomerID),
    CONSTRAINT fk_td_salesperson FOREIGN KEY (SalespersonID)
REFERENCES Salesperson(SalespersonID)
);

```

## 6. ShowRoom

```
CREATE TABLE Showroom
(
    ShowroomID NUMBER CONSTRAINT pk_showroom PRIMARY KEY,
    Name VARCHAR2(30) CONSTRAINT nn_showroom_name NOT NULL,
    Location VARCHAR2(50) CONSTRAINT nn_showroom_location NOT NULL,
    Phone VARCHAR2(15) CONSTRAINT nn_showroom_phone NOT NULL,
    AdminID NUMBER,
    CONSTRAINT fk_showroom_admin FOREIGN KEY (AdminID) REFERENCES
Administration(AdminID)
);
```

## 7. Inventory

```
CREATE TABLE Inventory
(
    InventoryID NUMBER CONSTRAINT pk_inventory PRIMARY KEY,
    ItemName VARCHAR2(25) CONSTRAINT nn_inventory_item NOT NULL,
    Quantity NUMBER CONSTRAINT chk_inventory_quantity CHECK (Quantity >=
0),
    Category VARCHAR2(20) CONSTRAINT nn_inventory_category NOT NULL,
    ShowroomID NUMBER,
    CONSTRAINT fk_inventory_showroom FOREIGN KEY (ShowroomID)
REFERENCES Showroom(ShowroomID)
);
```

## 8. Staff

```
CREATE TABLE Staff
(
    StaffID NUMBER CONSTRAINT pk_staff PRIMARY KEY,
    Name VARCHAR2(30) CONSTRAINT nn_staff_name NOT NULL,
    Role VARCHAR2(20) CONSTRAINT nn_staff_role NOT NULL,
    Contact VARCHAR2(15) CONSTRAINT nn_staff_contact NOT NULL,
    ShowroomID NUMBER,
    CONSTRAINT fk_staff_showroom FOREIGN KEY (ShowroomID)
REFERENCES Showroom(ShowroomID)
);
```

## 9. Payroll

```
CREATE TABLE Payroll
(
```

```

PayID NUMBER CONSTRAINT pk_payroll PRIMARY KEY,
SalaryAmount NUMBER,
PayDate DATE DEFAULT SYSDATE,
StaffID NUMBER,
CONSTRAINT fk_payroll_staff FOREIGN KEY (StaffID) REFERENCES
Staff(StaffID)
);

```

## 10. Administration

```

CREATE TABLE Administration
(
AdminID NUMBER CONSTRAINT pk_admin PRIMARY KEY,
Name VARCHAR2(30) CONSTRAINT nn_admin_name NOT NULL,
Contact VARCHAR2(15) CONSTRAINT nn_admin_contact NOT NULL,
Role VARCHAR2(20) CONSTRAINT nn_admin_role NOT NULL,
ShowroomID NUMBER,
CONSTRAINT fk_admin_showroom FOREIGN KEY (ShowroomID)
REFERENCES Showroom(ShowroomID)
);

```

## 11. Insurance

```

CREATE TABLE Insurance
(
InsuranceID NUMBER CONSTRAINT pk_insurance PRIMARY KEY,
InsuranceProvider VARCHAR2(30) CONSTRAINT nn_insurance_provider
NOT NULL,
InsuranceStartDate DATE,
InsuranceEndDate DATE,
CoverageAmount NUMBER,
CarID NUMBER,
CustomerID NUMBER,
CONSTRAINT fk_insurance_car FOREIGN KEY (CarID) REFERENCES
Car(CarID),
CONSTRAINT fk_insurance_customer FOREIGN KEY (CustomerID)
REFERENCES Customer(CustomerID)
);

```

## 12. Booking

```

CREATE TABLE Booking
(
BookingID NUMBER CONSTRAINT pk_booking PRIMARY KEY,

```

```

BookingDate DATE DEFAULT SYSDATE,
BookingStatus VARCHAR2(20) CONSTRAINT chk_booking_status CHECK
(BookingStatus IN ('Pending', 'Confirmed', 'Cancelled')),
AdvanceAmount NUMBER,
CustomerID NUMBER,
CarID NUMBER,
CONSTRAINT fk_booking_customer FOREIGN KEY (CustomerID)
REFERENCES Customer(CustomerID),
CONSTRAINT fk_booking_car FOREIGN KEY (CarID) REFERENCES
Car(CarID)
);

```

### 13. Warranty

```

CREATE TABLE Warranty
(
WarrantyID NUMBER CONSTRAINT pk_warranty PRIMARY KEY,
WarrantyStartDate DATE,
WarrantyEndDate DATE,
CoverageDetails VARCHAR2(100),
CarID NUMBER,
CONSTRAINT fk_warranty_car FOREIGN KEY (CarID) REFERENCES
Car(CarID)
);

```

## 2. Alter Statement

### 1. Car

```

ALTER TABLE Car
ADD CONSTRAINT chk_car_year CHECK (Year >= 1990 AND Year <=
EXTRACT(YEAR FROM SYSDATE));

```

### 2. Customer

```

ALTER TABLE Customer
ADD CONSTRAINT chk_customer_phone_length CHECK (LENGTH(Phone)
>= 10);

```

### 3. Salesperson

```

ALTER TABLE Salesperson
MODIFY Salary NUMBER DEFAULT 30000;

```

#### 4. Sale

```
ALTER TABLE Sale  
ADD CONSTRAINT fk_sale_customer FOREIGN KEY (CustomerID)  
REFERENCES Customer(CustomerID);
```

#### 5. TestDrive

```
ALTER TABLE TestDrive  
ADD CONSTRAINT chk_testdrive_duration CHECK  
(TO_NUMBER(Duration) BETWEEN 15 AND 180);
```

#### 6. ShowRoom

```
ALTER TABLE Showroom  
ADD CONSTRAINT uq_showroom_phone UNIQUE (Phone);
```

#### 7. Inventory

```
ALTER TABLE Inventory  
ADD CONSTRAINT fk_inventory_showroom FOREIGN KEY (ShowroomID)  
REFERENCES Showroom(ShowroomID);
```

#### 8. Staff

```
ALTER TABLE Staff  
MODIFY Role VARCHAR2(30) CONSTRAINT nn_staff_role NOT NULL;
```

#### 9. Payroll

```
ALTER TABLE Payroll  
ADD CONSTRAINT chk_payroll_salary CHECK (SalaryAmount >= 10000);
```

#### 10. Administration

```
ALTER TABLE Administration  
ADD CONSTRAINT uq_admin_contact UNIQUE (Contact);
```

#### 11. Insurance

```
ALTER TABLE Insurance  
ADD CONSTRAINT chk_insurance_dates CHECK (InsuranceEndDate >  
InsuranceStartDate);
```

## 12. Booking

```
ALTER TABLE Booking  
MODIFY BookingStatus VARCHAR2(20) CONSTRAINT nn_booking_status  
NOT NULL;
```

## 13. Warranty

```
ALTER TABLE Warranty  
ADD CONSTRAINT chk_warranty_dates CHECK (WarrantyEndDate >  
WarrantyStartDate);
```

# 3. Insert Statement

## 1. Customer

```
INSERT ALL  
INTO Customer VALUES (1, 'Ali Raza', '03001234567', 'ali@gmail.com',  
'Lahore')  
INTO Customer VALUES (2, 'Sara Khan', '03111234567',  
'sara@gmail.com', 'Karachi')  
INTO Customer VALUES (3, 'Bilal Ahmed', '03211234567',  
'bilal@gmail.com', 'Islamabad')  
INTO Customer VALUES (4, 'Zainab Fatima', '03321234567',  
'zainab@gmail.com', 'Faisalabad')  
INTO Customer VALUES (5, 'Usman Tariq', '03431234567',  
'usman@gmail.com', 'Multan')  
INTO Customer VALUES (6, 'Hamza Qureshi', '03541234567',  
'hamza@gmail.com', 'Rawalpindi')  
INTO Customer VALUES (7, 'Noor Jahan', '03651234567',  
'noor@gmail.com', 'Peshawar')  
INTO Customer VALUES (8, 'Fahad Aslam', '03761234567',  
'fahad@gmail.com', 'Quetta')  
INTO Customer VALUES (9, 'Hira Sheikh', '03871234567',  
'hira@gmail.com', 'Sialkot')  
INTO Customer VALUES (10, 'Adeel Latif', '03981234567',  
'adeel@gmail.com', 'Gujranwala')  
SELECT * FROM dual;
```

## 2. Car

```
INSERT ALL  
INTO Car VALUES (1, 'Toyota', 'Corolla', 2022, 'White', 'Petrol',  
'Automatic', 12000, 3200000, 'VIN001', 'Available')
```

```

INTO Car VALUES (2, 'Honda', 'Civic', 2021, 'Black', 'Petrol', 'Manual',
15000, 3500000, 'VIN002', 'Available')
INTO Car VALUES (3, 'Suzuki', 'Alto', 2023, 'Red', 'Petrol', 'Automatic',
5000, 1800000, 'VIN003', 'Available')
INTO Car VALUES (4, 'Kia', 'Sportage', 2020, 'Blue', 'Diesel', 'Automatic',
22000, 4500000, 'VIN004', 'Available')
INTO Car VALUES (5, 'Hyundai', 'Tucson', 2022, 'Grey', 'Petrol',
'Automatic', 10000, 4700000, 'VIN005', 'Available')
INTO Car VALUES (6, 'Nissan', 'Sunny', 2021, 'Silver', 'Petrol', 'Manual',
17000, 2800000, 'VIN006', 'Available')
INTO Car VALUES (7, 'MG', 'HS', 2023, 'White', 'Petrol', 'Automatic',
2000, 5200000, 'VIN007', 'Available')
INTO Car VALUES (8, 'Audi', 'A4', 2020, 'Black', 'Petrol', 'Automatic',
30000, 7500000, 'VIN008', 'Available')
INTO Car VALUES (9, 'BMW', 'X1', 2019, 'Blue', 'Diesel', 'Automatic',
40000, 8000000, 'VIN009', 'Available')
INTO Car VALUES (10, 'Changan', 'Alsvin', 2022, 'Red', 'Petrol',
'Manual', 7000, 2300000, 'VIN010', 'Available')
SELECT * FROM dual;

```

### 3. Salesperson

```

INSERT ALL
INTO Salesperson VALUES (1, 'Ahmed Iqbal', '03001111222',
'ahmed@sales.com', TO_DATE('2021-01-10','YYYY-MM-DD'), 'Senior',
60000)
INTO Salesperson VALUES (2, 'Laiba Noor', '03002222333',
'laiba@sales.com', TO_DATE('2022-03-15','YYYY-MM-DD'), 'Junior',
40000)
INTO Salesperson VALUES (3, 'Zeeshan Shah', '03003333444',
'zeeshan@sales.com', TO_DATE('2020-08-05','YYYY-MM-DD'), 'Senior',
62000)
INTO Salesperson VALUES (4, 'Anam Ali', '03004444555',
'anam@sales.com', TO_DATE('2021-07-18','YYYY-MM-DD'), 'Junior',
39000)
INTO Salesperson VALUES (5, 'Usama Saeed', '03005555666',
'usama@sales.com', TO_DATE('2019-06-25','YYYY-MM-DD'), 'Senior',
70000)
INTO Salesperson VALUES (6, 'Hassan Raza', '03006666777',
'hassan@sales.com', TO_DATE('2023-02-01','YYYY-MM-DD'), 'Trainee',
25000)

```

```

INTO Salesperson VALUES (7, 'Zoya Imran', '03007777888',
'zoya@sales.com', TO_DATE('2022-12-10','YYYY-MM-DD'), 'Junior',
41000)
INTO Salesperson VALUES (8, 'Ali Shan', '03008888999',
'ali@sales.com', TO_DATE('2020-11-11','YYYY-MM-DD'), 'Senior',
64000)
INTO Salesperson VALUES (9, 'Maira Khan', '03009999000',
'maira@sales.com', TO_DATE('2023-04-09','YYYY-MM-DD'), 'Trainee',
27000)
INTO Salesperson VALUES (10, 'Raza Malik', '03001010101',
'raza@sales.com', TO_DATE('2021-09-21','YYYY-MM-DD'), 'Junior',
42000)
SELECT * FROM dual;

```

#### 4. Sale

```

INSERT ALL
INTO Sale VALUES (1, TO_DATE('2025-06-26','YYYY-MM-DD'),
5828262, 'Credit Card', 1, 1, 1)
INTO Sale VALUES (2, TO_DATE('2025-06-19','YYYY-MM-DD'),
6341570, 'Credit Card', 2, 2, 2)
INTO Sale VALUES (3, TO_DATE('2025-06-03','YYYY-MM-DD'),
7426387, 'Cash', 3, 3, 3)
INTO Sale VALUES (4, TO_DATE('2025-06-01','YYYY-MM-DD'),
5229380, 'Cash', 4, 4, 4)
INTO Sale VALUES (5, TO_DATE('2025-06-18','YYYY-MM-DD'),
7834873, 'Cash', 5, 5, 5)
INTO Sale VALUES (6, TO_DATE('2025-06-01','YYYY-MM-DD'),
3576735, 'Credit Card', 6, 6, 6)
INTO Sale VALUES (7, TO_DATE('2025-06-17','YYYY-MM-DD'),
4112847, 'Cheque', 7, 7, 7)
INTO Sale VALUES (8, TO_DATE('2025-06-28','YYYY-MM-DD'),
7432774, 'Cash', 8, 8, 8)
INTO Sale VALUES (9, TO_DATE('2025-06-22','YYYY-MM-DD'),
2638714, 'Cash', 9, 9, 9)
INTO Sale VALUES (10, TO_DATE('2025-06-19','YYYY-MM-DD'),
4603626, 'Cheque', 10, 10, 10)
SELECT * FROM dual;

```

#### 5. TestDrive

```

INSERT ALL
INTO TestDrive VALUES (1, TO_DATE('2025-06-22','YYYY-MM-DD'),
'29', 'Smooth ride', 1, 1, 1)

```



```

INTO TestDrive VALUES (2, TO_DATE('2025-06-11','YYYY-MM-DD'),
'48', 'Smooth ride', 2, 2, 2)
INTO TestDrive VALUES (3, TO_DATE('2025-06-17','YYYY-MM-DD'),
'58', 'Smooth ride', 3, 3, 3)
INTO TestDrive VALUES (4, TO_DATE('2025-06-15','YYYY-MM-DD'),
'39', 'Smooth ride', 4, 4, 4)
INTO TestDrive VALUES (5, TO_DATE('2025-06-15','YYYY-MM-DD'),
'29', 'Smooth ride', 5, 5, 5)
INTO TestDrive VALUES (6, TO_DATE('2025-06-04','YYYY-MM-DD'),
'49', 'Smooth ride', 6, 6, 6)
INTO TestDrive VALUES (7, TO_DATE('2025-06-14','YYYY-MM-DD'),
'29', 'Smooth ride', 7, 7, 7)
INTO TestDrive VALUES (8, TO_DATE('2025-06-07','YYYY-MM-DD'),
'60', 'Smooth ride', 8, 8, 8)
INTO TestDrive VALUES (9, TO_DATE('2025-06-06','YYYY-MM-DD'),
'49', 'Smooth ride', 9, 9, 9)
INTO TestDrive VALUES (10, TO_DATE('2025-06-21','YYYY-MM-DD'),
'54', 'Smooth ride', 10, 10, 10)
SELECT * FROM dual;

```

## 6. ShowRoom

```

INSERT ALL
INTO Showroom VALUES (1, 'Showroom 1', 'Faisalabad', '0426471268',
1)
INTO Showroom VALUES (2, 'Showroom 2', 'Faisalabad', '0422546986',
2)
INTO Showroom VALUES (3, 'Showroom 3', 'Islamabad', '0426345720', 3)
INTO Showroom VALUES (4, 'Showroom 4', 'Faisalabad', '0426544286',
4)
INTO Showroom VALUES (5, 'Showroom 5', 'Multan', '0422509023', 5)
INTO Showroom VALUES (6, 'Showroom 6', 'Islamabad', '0423319795', 6)
INTO Showroom VALUES (7, 'Showroom 7', 'Karachi', '0427368487', 7)
INTO Showroom VALUES (8, 'Showroom 8', 'Faisalabad', '0422900660',
8)
INTO Showroom VALUES (9, 'Showroom 9', 'Islamabad', '0422930512', 9)
INTO Showroom VALUES (10, 'Showroom 10', 'Karachi', '0427584983',
10)
SELECT * FROM dual;

```

## 7. Inventory

```

INSERT ALL
INTO Inventory VALUES (1, 'Tires', 94, 'Accessory', 1)
INTO Inventory VALUES (2, 'Oil', 72, 'Maintenance', 2)

```

```

INTO Inventory VALUES (3, 'Cleaning Kit', 73, 'Accessory', 3)
INTO Inventory VALUES (4, 'Spare Battery', 74, 'Accessory', 4)
INTO Inventory VALUES (5, 'Brake Pads', 68, 'Accessory', 5)
INTO Inventory VALUES (6, 'AC Gas', 28, 'Accessory', 6)
INTO Inventory VALUES (7, 'Engine Oil', 23, 'Accessory', 7)
INTO Inventory VALUES (8, 'Wipers', 98, 'Maintenance', 8)
INTO Inventory VALUES (9, 'Headlights', 52, 'Maintenance', 9)
INTO Inventory VALUES (10, 'Floor Mats', 68, 'Maintenance', 10)
SELECT * FROM dual;

```

## 8. Staff

```

INSERT ALL
INTO Staff VALUES (1, 'Staff 1', 'Cleaner', '03982596877', 1)
INTO Staff VALUES (2, 'Staff 2', 'Cleaner', '03171536825', 2)
INTO Staff VALUES (3, 'Staff 3', 'Receptionist', '03238117948', 3)
INTO Staff VALUES (4, 'Staff 4', 'Receptionist', '03138884776', 4)
INTO Staff VALUES (5, 'Staff 5', 'Receptionist', '03749839315', 5)
INTO Staff VALUES (6, 'Staff 6', 'Mechanic', '03674006065', 6)
INTO Staff VALUES (7, 'Staff 7', 'Mechanic', '03513306731', 7)
INTO Staff VALUES (8, 'Staff 8', 'Receptionist', '03665407628', 8)
INTO Staff VALUES (9, 'Staff 9', 'Mechanic', '03255765771', 9)
INTO Staff VALUES (10, 'Staff 10', 'Mechanic', '03658576489', 10)
SELECT * FROM dual;

```

## 9. Payroll

```

INSERT ALL
INTO Payroll VALUES (1, 23286, TO_DATE('2025-06-28', 'YYYY-MM-DD'), 1)
INTO Payroll VALUES (2, 44724, TO_DATE('2025-06-20', 'YYYY-MM-DD'), 2)
INTO Payroll VALUES (3, 27426, TO_DATE('2025-06-17', 'YYYY-MM-DD'), 3)
INTO Payroll VALUES (4, 21278, TO_DATE('2025-06-14', 'YYYY-MM-DD'), 4)
INTO Payroll VALUES (5, 46908, TO_DATE('2025-06-18', 'YYYY-MM-DD'), 5)
INTO Payroll VALUES (6, 58685, TO_DATE('2025-06-02', 'YYYY-MM-DD'), 6)
INTO Payroll VALUES (7, 23262, TO_DATE('2025-06-18', 'YYYY-MM-DD'), 7)
INTO Payroll VALUES (8, 35874, TO_DATE('2025-06-14', 'YYYY-MM-DD'), 8)

```

```

INTO Payroll VALUES (9, 57895, TO_DATE('2025-06-01', 'YYYY-MM-DD'), 9)
INTO Payroll VALUES (10, 42732, TO_DATE('2025-06-15', 'YYYY-MM-DD'), 10)
SELECT * FROM dual;

```

## 10. Administration

```

INSERT ALL
INTO Administration VALUES (1, 'Admin 1', '03395894313', 'Manager',
1)
INTO Administration VALUES (2, 'Admin 2', '03659109864',
'Coordinator', 2)
INTO Administration VALUES (3, 'Admin 3', '03995320967', 'Executive',
3)
INTO Administration VALUES (4, 'Admin 4', '03368527967',
'Coordinator', 4)
INTO Administration VALUES (5, 'Admin 5', '03628034134', 'Manager',
5)
INTO Administration VALUES (6, 'Admin 6', '03428566305', 'Executive',
6)
INTO Administration VALUES (7, 'Admin 7', '03957675188', 'Supervisor',
7)
INTO Administration VALUES (8, 'Admin 8', '03169264814', 'Executive',
8)
INTO Administration VALUES (9, 'Admin 9', '03210513403', 'Manager',
9)
INTO Administration VALUES (10, 'Admin 10', '03085395621',
'Coordinator', 10)
SELECT * FROM dual;

```

## 11. Insurance

```

INSERT ALL
INTO Insurance VALUES (1, 'Company 1', TO_DATE('2025-01-01', 'YYYY-MM-DD'), TO_DATE('2026-01-01', 'YYYY-MM-DD'), 118446, 1, 1)
INTO Insurance VALUES (2, 'Company 2', TO_DATE('2025-01-01', 'YYYY-MM-DD'), TO_DATE('2026-01-01', 'YYYY-MM-DD'), 382339, 2, 2)
INTO Insurance VALUES (3, 'Company 3', TO_DATE('2025-01-01', 'YYYY-MM-DD'), TO_DATE('2026-01-01', 'YYYY-MM-DD'), 418731, 3, 3)
INTO Insurance VALUES (4, 'Company 4', TO_DATE('2025-01-01', 'YYYY-MM-DD'), TO_DATE('2026-01-01', 'YYYY-MM-DD'), 172474, 4, 4)
INTO Insurance VALUES (5, 'Company 5', TO_DATE('2025-01-01', 'YYYY-MM-DD'), TO_DATE('2026-01-01', 'YYYY-MM-DD'), 232923, 5, 5)

```

```

INTO Insurance VALUES (6, 'Company 6', TO_DATE('2025-01-01','YYYY-MM-DD'), TO_DATE('2026-01-01','YYYY-MM-DD'), 213089, 6, 6)
INTO Insurance VALUES (7, 'Company 7', TO_DATE('2025-01-01','YYYY-MM-DD'), TO_DATE('2026-01-01','YYYY-MM-DD'), 352404, 7, 7)
INTO Insurance VALUES (8, 'Company 8', TO_DATE('2025-01-01','YYYY-MM-DD'), TO_DATE('2026-01-01','YYYY-MM-DD'), 120058, 8, 8)
INTO Insurance VALUES (9, 'Company 9', TO_DATE('2025-01-01','YYYY-MM-DD'), TO_DATE('2026-01-01','YYYY-MM-DD'), 367928, 9, 9)
INTO Insurance VALUES (10, 'Company 10', TO_DATE('2025-01-01','YYYY-MM-DD'), TO_DATE('2026-01-01','YYYY-MM-DD'), 483809, 10, 10)
SELECT * FROM dual;

```

## 12. Booking

```

INSERT ALL
INTO Booking VALUES (1, TO_DATE('2025-06-03','YYYY-MM-DD'), 'Cancelled', 124575, 1, 1)
INTO Booking VALUES (2, TO_DATE('2025-06-04','YYYY-MM-DD'), 'Confirmed', 136418, 2, 2)
INTO Booking VALUES (3, TO_DATE('2025-06-19','YYYY-MM-DD'), 'Pending', 186487, 3, 3)
INTO Booking VALUES (4, TO_DATE('2025-06-05','YYYY-MM-DD'), 'Pending', 138030, 4, 4)
INTO Booking VALUES (5, TO_DATE('2025-06-15','YYYY-MM-DD'), 'Pending', 64283, 5, 5)
INTO Booking VALUES (6, TO_DATE('2025-06-01','YYYY-MM-DD'), 'Cancelled', 78664, 6, 6)
INTO Booking VALUES (7, TO_DATE('2025-06-20','YYYY-MM-DD'), 'Cancelled', 117273, 7, 7)
INTO Booking VALUES (8, TO_DATE('2025-06-05','YYYY-MM-DD'), 'Pending', 64087, 8, 8)
INTO Booking VALUES (9, TO_DATE('2025-06-24','YYYY-MM-DD'), 'Confirmed', 199373, 9, 9)
INTO Booking VALUES (10, TO_DATE('2025-06-08','YYYY-MM-DD'), 'Cancelled', 75096, 10, 10)
SELECT * FROM dual;

```

## 13. Warranty

```

INSERT ALL
INTO Warranty VALUES (1, TO_DATE('2025-01-01','YYYY-MM-DD'), TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year', 1)

```

```

INTO Warranty VALUES (2, TO_DATE('2025-01-01','YYYY-MM-DD'),
TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year',
2)
INTO Warranty VALUES (3, TO_DATE('2025-01-01','YYYY-MM-DD'),
TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year',
3)
INTO Warranty VALUES (4, TO_DATE('2025-01-01','YYYY-MM-DD'),
TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year',
4)
INTO Warranty VALUES (5, TO_DATE('2025-01-01','YYYY-MM-DD'),
TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year',
5)
INTO Warranty VALUES (6, TO_DATE('2025-01-01','YYYY-MM-DD'),
TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year',
6)
INTO Warranty VALUES (7, TO_DATE('2025-01-01','YYYY-MM-DD'),
TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year',
7)
INTO Warranty VALUES (8, TO_DATE('2025-01-01','YYYY-MM-DD'),
TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year',
8)
INTO Warranty VALUES (9, TO_DATE('2025-01-01','YYYY-MM-DD'),
TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year',
9)
INTO Warranty VALUES (10, TO_DATE('2025-01-01','YYYY-MM-DD'),
TO_DATE('2026-01-01','YYYY-MM-DD'), 'Standard coverage for 1 year',
10)
SELECT * FROM dual;

```

## *(Views Creation)*

### 1. *AvailableCarsView*

```

CREATE VIEW AvailableCarsView AS
SELECT CarID, Make, Model, Year, Price, Color
FROM Car
WHERE Status = 'Available';

```

| CARID | MAKE    | MODEL    | YEAR | PRICE   | COLOR  |
|-------|---------|----------|------|---------|--------|
| 1     | Toyota  | Corolla  | 2022 | 3200000 | White  |
| 2     | Honda   | Civic    | 2021 | 3500000 | Black  |
| 3     | Suzuki  | Alto     | 2023 | 1800000 | Red    |
| 4     | Kia     | Sportage | 2020 | 4500000 | Blue   |
| 5     | Hyundai | Tucson   | 2022 | 4700000 | Grey   |
| 6     | Nissan  | Sunny    | 2021 | 2800000 | Silver |
| 7     | MG      | HS       | 2023 | 5200000 | White  |
| 8     | Audi    | A4       | 2020 | 7500000 | Black  |
| 9     | BMW     | X1       | 2019 | 8000000 | Blue   |
| 10    | Changan | Alsvin   | 2022 | 2300000 | Red    |

10 rows returned in 0.01 seconds

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## 2. SalesReportView

```

CREATE VIEW SalesReportView AS
SELECT
    S.SaleID,
    C.Name AS CustomerName,
    Car.Make || ' ' || Car.Model AS Car,
    S.FinalPrice,
    S.PaymentMethod,
    S.DateOfSale
FROM Sale S
JOIN Customer C ON S.CustomerID = C.CustomerID
JOIN Car ON S.CarID = Car.CarID;

```

| SALEID | CUSTOMERNAME  | CAR            | FINALPRICE | PAYMENTMETHOD | DATEOFSALE |
|--------|---------------|----------------|------------|---------------|------------|
| 1      | Ali Raza      | Toyota Corolla | 5828262    | Credit Card   | 06/26/2025 |
| 2      | Sara Khan     | Honda Civic    | 6341570    | Credit Card   | 06/19/2025 |
| 3      | Bilal Ahmed   | Suzuki Alto    | 7426387    | Cash          | 06/03/2025 |
| 4      | Zainab Fatima | Kia Sportage   | 5229380    | Cash          | 06/01/2025 |
| 5      | Usman Tariq   | Hyundai Tucson | 7834873    | Cash          | 06/18/2025 |
| 6      | Hamza Qureshi | Nissan Sunny   | 3576735    | Credit Card   | 06/01/2025 |
| 7      | Noor Jahan    | MG HS          | 4112847    | Cheque        | 06/17/2025 |
| 8      | Fahad Aslam   | Audi A4        | 7432774    | Cash          | 06/28/2025 |
| 9      | Hira Sheikh   | BMW X1         | 2638714    | Cash          | 06/22/2025 |
| 10     | Adeel Latif   | Changan Alsvin | 4603626    | Cheque        | 06/19/2025 |

10 rows returned in 0.01 seconds

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## 3. BookingStatusView

```

CREATE VIEW BookingStatusView AS
SELECT
    B.BookingID,
    C.Name AS CustomerName,

```

```

Car.Make || ' ' || Car.Model AS Car,
B.BookingDate,
B.BookingStatus,
B.AdvanceAmount
FROM Booking B
JOIN Customer C ON B.CustomerID = C.CustomerID
JOIN Car ON B.CarID = Car.CarID;

```

| BOOKINGID | CUSTOMERNAME  | CAR            | BOOKINGDATE | BOOKINGSTATUS | ADVANCEAMOUNT |
|-----------|---------------|----------------|-------------|---------------|---------------|
| 1         | Ali Raza      | Toyota Corolla | 06/03/2025  | Cancelled     | 124575        |
| 2         | Sara Khan     | Honda Civic    | 06/04/2025  | Confirmed     | 136418        |
| 3         | Bilal Ahmed   | Suzuki Alto    | 06/19/2025  | Pending       | 186487        |
| 4         | Zainab Fatima | Kia Sportage   | 06/05/2025  | Pending       | 138030        |
| 5         | Usman Tariq   | Hyundai Tucson | 06/15/2025  | Pending       | 64283         |
| 6         | Hamza Qureshi | Nissan Sunny   | 06/01/2025  | Cancelled     | 78664         |
| 7         | Noor Jahan    | MG HS          | 06/20/2025  | Cancelled     | 117273        |
| 8         | Fahad Aslam   | Audi A4        | 06/05/2025  | Pending       | 64087         |
| 9         | Hira Sheikh   | BMW X1         | 06/24/2025  | Confirmed     | 199373        |
| 10        | Adeel Latif   | Changan Alsvin | 06/08/2025  | Cancelled     | 75096         |

10 rows returned in 0.01 seconds

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#### 4. TestDriveScheduleView

```

CREATE VIEW TestDriveScheduleView AS
SELECT
    TD.TestDriveID,
    SP.Name AS SalespersonName,
    C.Name AS CustomerName,
    Car.Make || ' ' || Car.Model AS Car,
    TD.ScheduledDate,
    TD.Duration,
    TD.Feedback
FROM TestDrive TD
JOIN Salesperson SP ON TD.SalespersonID = SP.SalespersonID
JOIN Customer C ON TD.CustomerID = C.CustomerID
JOIN Car ON TD.CarID = Car.CarID;

```

| TESTDRIVEID | SALESPERSONNAME | CUSTOMERNAME  | CAR            | SCHEDULEDDATE | DURATION | FEEDBACK    |
|-------------|-----------------|---------------|----------------|---------------|----------|-------------|
| 1           | Ahmed Iqbal     | Ali Raza      | Toyota Corolla | 06/22/2025    | 29       | Smooth ride |
| 2           | Laiba Noor      | Sara Khan     | Honda Civic    | 06/11/2025    | 48       | Smooth ride |
| 3           | Zeeshan Shah    | Bilal Ahmed   | Suzuki Alto    | 06/17/2025    | 58       | Smooth ride |
| 4           | Anam Ali        | Zainab Fatima | Kia Sportage   | 06/15/2025    | 39       | Smooth ride |
| 5           | Usama Saeed     | Usman Tariq   | Hyundai Tucson | 06/15/2025    | 29       | Smooth ride |
| 6           | Hassan Raza     | Hamza Qureshi | Nissan Sunny   | 06/04/2025    | 49       | Smooth ride |
| 7           | Zoya Imran      | Noor Jahan    | MG HS          | 06/14/2025    | 29       | Smooth ride |
| 8           | Ali Shan        | Fahad Aslam   | Audi A4        | 06/07/2025    | 60       | Smooth ride |
| 9           | Maira Khan      | Hira Sheikh   | BMW X1         | 06/06/2025    | 49       | Smooth ride |
| 10          | Raza Malik      | Adeel Latif   | Changan Alsvin | 06/21/2025    | 54       | Smooth ride |

10 rows returned in 0.01 seconds

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## 5. ShowroomStaffView

```
CREATE VIEW ShowroomStaffView AS
SELECT
    S.StaffID,
    S.Name AS StaffName,
    S.Role,
    SR.Name AS ShowroomName,
    SR.Location
FROM Staff S
JOIN Showroom SR ON S.ShowroomID = SR.ShowroomID;
```

| STAFFID | STAFFNAME | ROLE         | SHOWROOMNAME | LOCATION   |
|---------|-----------|--------------|--------------|------------|
| 1       | Staff 1   | Cleaner      | Showroom 1   | Faisalabad |
| 2       | Staff 2   | Cleaner      | Showroom 2   | Faisalabad |
| 3       | Staff 3   | Receptionist | Showroom 3   | Islamabad  |
| 4       | Staff 4   | Receptionist | Showroom 4   | Faisalabad |
| 5       | Staff 5   | Receptionist | Showroom 5   | Multan     |
| 6       | Staff 6   | Mechanic     | Showroom 6   | Islamabad  |
| 7       | Staff 7   | Mechanic     | Showroom 7   | Karachi    |
| 8       | Staff 8   | Receptionist | Showroom 8   | Faisalabad |
| 9       | Staff 9   | Mechanic     | Showroom 9   | Islamabad  |
| 10      | Staff 10  | Mechanic     | Showroom 10  | Karachi    |

10 rows returned in 0.01 seconds      [Download](#)

# (PL/SQL Statements)

## 1. Auto-update Car Status to 'Sold' after a Sale

```
CREATE OR REPLACE TRIGGER trg_update_car_status_after_sale
AFTER INSERT ON Sale
FOR EACH ROW
BEGIN
    UPDATE Car
    SET Status = 'Sold'
    WHERE CarID = :NEW.CarID;
END;
/
```



## **2. Calculate Total Sales for a Salesperson**

```
CREATE OR REPLACE FUNCTION fn_total_sales_by_salesperson(sp_id NUMBER)
RETURN NUMBER
IS
    total_sales NUMBER := 0;
BEGIN
    SELECT SUM(FinalPrice)
    INTO total_sales
    FROM Sale
    WHERE SalespersonID = sp_id;

    RETURN NVL(total_sales, 0);
END;
/
```

## **3. Confirm a Booking**

```
CREATE OR REPLACE PROCEDURE pr_confirm_booking(p_booking_id IN NUMBER)
IS
BEGIN
    UPDATE Booking
    SET BookingStatus = 'Confirmed'
    WHERE BookingID = p_booking_id;

    DBMS_OUTPUT.PUT_LINE('Booking ' || p_booking_id || ' confirmed.');
```

```
END;
/
```

## **4. Prevent Duplicate VINs**

```
CREATE OR REPLACE TRIGGER trg_prevent_duplicate_vin
BEFORE INSERT OR UPDATE ON Car
FOR EACH ROW
DECLARE
    v_count NUMBER;
BEGIN
    SELECT COUNT(*) INTO v_count
    FROM Car
    WHERE VIN = :NEW.VIN AND CarID != :NEW.CarID;

    IF v_count > 0 THEN
        RAISE_APPLICATION_ERROR(-20001, 'Duplicate VIN not allowed.');
```

```
END IF;
```

```
END;
/
```

## 5. Validate Insurance Dates

```
CREATE OR REPLACE TRIGGER trg_validate_insurance_dates
BEFORE INSERT OR UPDATE ON Insurance
FOR EACH ROW
BEGIN
    IF :NEW.InsuranceEndDate <= :NEW.InsuranceStartDate THEN
        RAISE_APPLICATION_ERROR(-20002, 'Insurance end date must be after start date.');
```

```
    END IF;
END;
/
```

## (Tables)

### 1. Car

| EDIT                                                                                | CARID | MAKE    | MODEL    | YEAR | COLOR  | ENGINE TYPE | TRANSMISSION | MILEAGE | PRICE   | VIN    | STATUS    |
|-------------------------------------------------------------------------------------|-------|---------|----------|------|--------|-------------|--------------|---------|---------|--------|-----------|
|  | 1     | Toyota  | Corolla  | 2022 | White  | Petrol      | Automatic    | 12000   | 3200000 | VIN001 | Available |
|  | 2     | Honda   | Civic    | 2021 | Black  | Petrol      | Manual       | 15000   | 3500000 | VIN002 | Available |
|  | 3     | Suzuki  | Alto     | 2023 | Red    | Petrol      | Automatic    | 5000    | 1800000 | VIN003 | Available |
|  | 4     | Kia     | Sportage | 2020 | Blue   | Diesel      | Automatic    | 22000   | 4500000 | VIN004 | Available |
|  | 5     | Hyundai | Tucson   | 2022 | Grey   | Petrol      | Automatic    | 10000   | 4700000 | VIN005 | Available |
|  | 6     | Nissan  | Sunny    | 2021 | Silver | Petrol      | Manual       | 17000   | 2800000 | VIN006 | Available |
|  | 7     | MG      | HS       | 2023 | White  | Petrol      | Automatic    | 2000    | 5200000 | VIN007 | Available |
|  | 8     | Audi    | A4       | 2020 | Black  | Petrol      | Automatic    | 30000   | 7500000 | VIN008 | Available |
|  | 9     | BMW     | X1       | 2019 | Blue   | Diesel      | Automatic    | 40000   | 8000000 | VIN009 | Available |
|  | 10    | Changan | Alsvin   | 2022 | Red    | Petrol      | Manual       | 7000    | 2300000 | VIN010 | Available |
| row(s) 1 - 10 of 10                                                                 |       |         |          |      |        |             |              |         |         |        |           |

## 2. Customer

| EDIT                                                                               | CUSTOMERID | NAME          | PHONE       | EMAIL            | ADDRESS    |
|------------------------------------------------------------------------------------|------------|---------------|-------------|------------------|------------|
|   | 1          | Ali Raza      | 03001234567 | ali@gmail.com    | Lahore     |
|   | 2          | Sara Khan     | 03111234567 | sara@gmail.com   | Karachi    |
|   | 3          | Bilal Ahmed   | 03211234567 | bilal@gmail.com  | Islamabad  |
|   | 4          | Zainab Fatima | 03321234567 | zainab@gmail.com | Faisalabad |
|   | 5          | Usman Tariq   | 03431234567 | usman@gmail.com  | Multan     |
|   | 6          | Hamza Qureshi | 03541234567 | hamza@gmail.com  | Rawalpindi |
|   | 7          | Noor Jahan    | 03651234567 | noor@gmail.com   | Peshawar   |
|   | 8          | Fahad Aslam   | 03761234567 | fahad@gmail.com  | Quetta     |
|   | 9          | Hira Sheikh   | 03871234567 | hira@gmail.com   | Sialkot    |
|  | 10         | Adeel Latif   | 03981234567 | adeel@gmail.com  | Gujranwala |
| row(s) 1 - 10 of 10                                                                |            |               |             |                  |            |

### 3. Salesperson

| EDIT                                                                                | SALESPERSONID | NAME         | PHONE       | EMAIL             | HIREDATE   | POSITION | SALARY |
|-------------------------------------------------------------------------------------|---------------|--------------|-------------|-------------------|------------|----------|--------|
|  | 1             | Ahmed Iqbal  | 03001111222 | ahmed@sales.com   | 01/10/2021 | Senior   | 60000  |
|  | 2             | Laiba Noor   | 03002222333 | laiba@sales.com   | 03/15/2022 | Junior   | 40000  |
|  | 3             | Zeeshan Shah | 03003333444 | zeeshan@sales.com | 08/05/2020 | Senior   | 62000  |
|  | 4             | Anam Ali     | 03004444555 | anam@sales.com    | 07/18/2021 | Junior   | 39000  |
|  | 5             | Usama Saeed  | 03005555666 | usama@sales.com   | 06/25/2019 | Senior   | 70000  |
|  | 6             | Hassan Raza  | 03006666777 | hassan@sales.com  | 02/01/2023 | Trainee  | 25000  |
|  | 7             | Zoya Imran   | 03007777888 | zoya@sales.com    | 12/10/2022 | Junior   | 41000  |
|  | 8             | Ali Shan     | 03008888999 | ali@sales.com     | 11/11/2020 | Senior   | 64000  |
|  | 9             | Maira Khan   | 03009999000 | maira@sales.com   | 04/09/2023 | Trainee  | 27000  |
|  | 10            | Raza Malik   | 03001010101 | raza@sales.com    | 09/21/2021 | Junior   | 42000  |
| row(s) 1 - 10 of 10                                                                 |               |              |             |                   |            |          |        |

#### **4. Sale**

| EDIT                | SALEID | DATEOFSALE | FINALPRICE | PAYMENTMETHOD | CUSTOMERID | CARID | SALESPERSONID |
|---------------------|--------|------------|------------|---------------|------------|-------|---------------|
|                     | 1      | 06/26/2025 | 5828262    | Credit Card   | 1          | 1     | 1             |
|                     | 2      | 06/19/2025 | 6341570    | Credit Card   | 2          | 2     | 2             |
|                     | 3      | 06/03/2025 | 7426387    | Cash          | 3          | 3     | 3             |
|                     | 4      | 06/01/2025 | 5229380    | Cash          | 4          | 4     | 4             |
|                     | 5      | 06/18/2025 | 7834873    | Cash          | 5          | 5     | 5             |
|                     | 6      | 06/01/2025 | 3576735    | Credit Card   | 6          | 6     | 6             |
|                     | 7      | 06/17/2025 | 4112847    | Cheque        | 7          | 7     | 7             |
|                     | 8      | 06/28/2025 | 7432774    | Cash          | 8          | 8     | 8             |
|                     | 9      | 06/22/2025 | 2638714    | Cash          | 9          | 9     | 9             |
|                     | 10     | 06/19/2025 | 4603626    | Cheque        | 10         | 10    | 10            |
| row(s) 1 - 10 of 10 |        |            |            |               |            |       |               |

## 5. TestDrive

| EDIT                                                                                | TESTDRIVEID | SCHEDULEDDATE | DURATION | FEEDBACK    | CARID | CUSTOMERID | SALESPERSONID |
|-------------------------------------------------------------------------------------|-------------|---------------|----------|-------------|-------|------------|---------------|
|  | 1           | 06/22/2025    | 29       | Smooth ride | 1     | 1          | 1             |
|  | 2           | 06/11/2025    | 48       | Smooth ride | 2     | 2          | 2             |
|  | 3           | 06/17/2025    | 58       | Smooth ride | 3     | 3          | 3             |
|  | 4           | 06/15/2025    | 39       | Smooth ride | 4     | 4          | 4             |
|  | 5           | 06/15/2025    | 29       | Smooth ride | 5     | 5          | 5             |
|  | 6           | 06/04/2025    | 49       | Smooth ride | 6     | 6          | 6             |
|  | 7           | 06/14/2025    | 29       | Smooth ride | 7     | 7          | 7             |
|  | 8           | 06/07/2025    | 60       | Smooth ride | 8     | 8          | 8             |
|  | 9           | 06/06/2025    | 49       | Smooth ride | 9     | 9          | 9             |
|  | 10          | 06/21/2025    | 54       | Smooth ride | 10    | 10         | 10            |
| row(s) 1 - 10 of 10                                                                 |             |               |          |             |       |            |               |

## 6. ShowRoom

| EDIT                                                                              | SHOWROOMID | NAME        | LOCATION   | PHONE      | ADMINID |
|-----------------------------------------------------------------------------------|------------|-------------|------------|------------|---------|
|  | 1          | Showroom 1  | Faisalabad | 0426471268 | 1       |
|  | 2          | Showroom 2  | Faisalabad | 0422546986 | 2       |
|  | 3          | Showroom 3  | Islamabad  | 0426345720 | 3       |
|  | 4          | Showroom 4  | Faisalabad | 0426544286 | 4       |
|  | 5          | Showroom 5  | Multan     | 0422509023 | 5       |
|  | 6          | Showroom 6  | Islamabad  | 0423319795 | 6       |
|  | 7          | Showroom 7  | Karachi    | 0427368487 | 7       |
|  | 8          | Showroom 8  | Faisalabad | 0422900660 | 8       |
|  | 9          | Showroom 9  | Islamabad  | 0422930512 | 9       |
|  | 10         | Showroom 10 | Karachi    | 0427584983 | 10      |
| row(s) 1 - 10 of 10                                                               |            |             |            |            |         |

## 7. Inventory

| EDIT                                                                                | INVENTORYID | ITEMNAME      | QUANTITY | CATEGORY    | SHOWROOMID |
|-------------------------------------------------------------------------------------|-------------|---------------|----------|-------------|------------|
|  | 1           | Tires         | 94       | Accessory   | 1          |
|  | 2           | Oil           | 72       | Maintenance | 2          |
|  | 3           | Cleaning Kit  | 73       | Accessory   | 3          |
|  | 4           | Spare Battery | 74       | Accessory   | 4          |
|  | 5           | Brake Pads    | 68       | Accessory   | 5          |
|  | 6           | AC Gas        | 28       | Accessory   | 6          |
|  | 7           | Engine Oil    | 23       | Accessory   | 7          |
|  | 8           | Wipers        | 98       | Maintenance | 8          |
|  | 9           | Headlights    | 52       | Maintenance | 9          |
|  | 10          | Floor Mats    | 68       | Maintenance | 10         |
| row(s) 1 - 10 of 10                                                                 |             |               |          |             |            |

## 8. Staff

| EDIT                                                                              | STAFFID | NAME     | ROLE         | CONTACT     | SHOWROOMID |
|-----------------------------------------------------------------------------------|---------|----------|--------------|-------------|------------|
|  | 1       | Staff 1  | Cleaner      | 03982596877 | 1          |
|  | 2       | Staff 2  | Cleaner      | 03171536825 | 2          |
|  | 3       | Staff 3  | Receptionist | 03238117948 | 3          |
|  | 4       | Staff 4  | Receptionist | 03138884776 | 4          |
|  | 5       | Staff 5  | Receptionist | 03749839315 | 5          |
|  | 6       | Staff 6  | Mechanic     | 03674006065 | 6          |
|  | 7       | Staff 7  | Mechanic     | 03513306731 | 7          |
|  | 8       | Staff 8  | Receptionist | 03665407628 | 8          |
|  | 9       | Staff 9  | Mechanic     | 03255765771 | 9          |
|  | 10      | Staff 10 | Mechanic     | 03658576489 | 10         |
| row(s) 1 - 10 of 10                                                               |         |          |              |             |            |

## 9. Payroll

| EDIT                                                                                | PAYID | SALARYAMOUNT | PAYDATE    | STAFFID |
|-------------------------------------------------------------------------------------|-------|--------------|------------|---------|
|  | 1     | 23286        | 06/28/2025 | 1       |
|  | 2     | 44724        | 06/20/2025 | 2       |
|  | 3     | 27426        | 06/17/2025 | 3       |
|  | 4     | 21278        | 06/14/2025 | 4       |
|  | 5     | 46908        | 06/18/2025 | 5       |
|  | 6     | 58685        | 06/02/2025 | 6       |
|  | 7     | 23262        | 06/18/2025 | 7       |
|  | 8     | 35874        | 06/14/2025 | 8       |
|  | 9     | 57895        | 06/01/2025 | 9       |
|  | 10    | 42732        | 06/15/2025 | 10      |
| row(s) 1 - 10 of 10                                                                 |       |              |            |         |

## 10. Administration

| EDIT                                                                              | ADMINID | NAME     | CONTACT     | ROLE        | SHOWROOMID |
|-----------------------------------------------------------------------------------|---------|----------|-------------|-------------|------------|
|  | 1       | Admin 1  | 03395894313 | Manager     | 10         |
|  | 2       | Admin 2  | 03659109864 | Coordinator | 2          |
|  | 3       | Admin 3  | 03995320967 | Executive   | 3          |
|  | 4       | Admin 4  | 03368527967 | Coordinator | 4          |
|  | 5       | Admin 5  | 03628034134 | Manager     | 5          |
|  | 6       | Admin 6  | 03428566305 | Executive   | 6          |
|  | 7       | Admin 7  | 03957675188 | Supervisor  | 7          |
|  | 8       | Admin 8  | 03169264814 | Executive   | 8          |
|  | 9       | Admin 9  | 03210513403 | Manager     | 9          |
|  | 10      | Admin 10 | 03085395621 | Coordinator | -          |
| row(s) 1 - 10 of 10                                                               |         |          |             |             |            |

## 11. Insurance

| EDIT                                                                                | INSURANCEID | INSURANCEPROVIDER | INSURANCESTARTDATE | INSURANCEENDDATE | COVERAGEAMOUNT | CARID | CUSTOMERID |
|-------------------------------------------------------------------------------------|-------------|-------------------|--------------------|------------------|----------------|-------|------------|
|  | 1           | Company 1         | 01/01/2025         | 01/01/2026       | 118446         | 1     | 1          |
|  | 2           | Company 2         | 01/01/2025         | 01/01/2026       | 382339         | 2     | 2          |
|  | 3           | Company 3         | 01/01/2025         | 01/01/2026       | 418731         | 3     | 3          |
|  | 4           | Company 4         | 01/01/2025         | 01/01/2026       | 172474         | 4     | 4          |
|  | 5           | Company 5         | 01/01/2025         | 01/01/2026       | 232923         | 5     | 5          |
|  | 6           | Company 6         | 01/01/2025         | 01/01/2026       | 213089         | 6     | 6          |
|  | 7           | Company 7         | 01/01/2025         | 01/01/2026       | 352404         | 7     | 7          |
|  | 8           | Company 8         | 01/01/2025         | 01/01/2026       | 120058         | 8     | 8          |
|  | 9           | Company 9         | 01/01/2025         | 01/01/2026       | 367928         | 9     | 9          |
|  | 10          | Company 10        | 01/01/2025         | 01/01/2026       | 483809         | 10    | 10         |
| row(s) 1 - 10 of 10                                                                 |             |                   |                    |                  |                |       |            |

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## 12. Booking

| EDIT                                                                              | BOOKINGID | BOOKINGDATE | BOOKINGSTATUS | ADVANCEAMOUNT | CUSTOMERID | CARID |
|-----------------------------------------------------------------------------------|-----------|-------------|---------------|---------------|------------|-------|
|  | 1         | 06/03/2025  | Cancelled     | 124575        | 1          | 1     |
|  | 2         | 06/04/2025  | Confirmed     | 136418        | 2          | 2     |
|  | 3         | 06/19/2025  | Pending       | 186487        | 3          | 3     |
|  | 4         | 06/05/2025  | Pending       | 138030        | 4          | 4     |
|  | 5         | 06/15/2025  | Pending       | 64283         | 5          | 5     |
|  | 6         | 06/01/2025  | Cancelled     | 78664         | 6          | 6     |
|  | 7         | 06/20/2025  | Cancelled     | 117273        | 7          | 7     |
|  | 8         | 06/05/2025  | Pending       | 64087         | 8          | 8     |
|  | 9         | 06/24/2025  | Confirmed     | 199373        | 9          | 9     |
|  | 10        | 06/08/2025  | Cancelled     | 75096         | 10         | 10    |
| row(s) 1 - 10 of 10                                                               |           |             |               |               |            |       |

## 13. Warranty

| EDIT                                                                                | WARRANTYID | WARRANTYSTARTDATE | WARRANTYENDDATE | COVERAGEDETAILS              | CARID |
|-------------------------------------------------------------------------------------|------------|-------------------|-----------------|------------------------------|-------|
|  | 1          | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 1     |
|  | 2          | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 2     |
|  | 3          | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 3     |
|  | 4          | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 4     |
|  | 5          | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 5     |
|  | 6          | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 6     |
|  | 7          | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 7     |
|  | 8          | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 8     |
|  | 9          | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 9     |
|  | 10         | 01/01/2025        | 01/01/2026      | Standard coverage for 1 year | 10    |
| row(s) 1 - 10 of 10                                                                 |            |                   |                 |                              |       |



